

Redaction Box CV 3.6

From scanned page to auditable geometry

A concise visual companion to the shipped classical computer-vision detector.

INPUT

One grayscale page at 200 DPI. No OCR, paired page, answer, document ID, Astra output, or manual label.

OUTPUT

Physical polygons plus reading-order groups. Geometry and semantic interpretation remain separate.

EVALUATION

1,414 targets over 1,277 page pairs; 0 page-pair errors.

The three exhibits are real source PDF pages and use the exact 3.6 detector.

The six-stage mental model

1. Page
Normalize one
raster page

2. Pixel masks
Separate dark and
faint evidence

3. Lines
Measure
horizontal and
vertical support

4. Proposals
Propose several
physical
explanations

5. Guards
Promote measured
geometry and
block artifacts

6. Consolidate
Freeze
components, then
infer reading
units

PHYSICAL LAYER

Stages 1-5 estimate page scale, generate competing geometric explanations, and reject text, glyph, margin, and scan artifacts. Candidate counts describe search volume, not predictions.

READING LAYER

Stage 6 freezes polygons before grouping. R2.1 and R2.2 can be one reading unit while remaining distinct boxes. Visible text is never filled to simplify the group.

WHY THIS ORDER MATTERS

A box can be physically valid yet belong to a different sentence or paragraph. Conversely, two adjacent components can form one redacted phrase. Separating geometry from reading order makes both decisions inspectable.

From proposal to emitted component

1. MEASURE

Deskew; estimate line height and content bounds; build dark/faint masks and line support.

2. PROPOSE

Open independent routes for outlines, blank contours, broken rails, steps, nesting, and blackouts.

3. SUPPORT

Require route-specific walls, corners, blankness, scale, or a page-level dense-ink regime.

4. REJECT

Remove glyphs, text slabs, scanner frames, margin clutter, unsupported walls, and duplicates.

5. FREEZE

Save physical polygons first; then group only by measured contact and defensible reading flow.

PROMOTION EXAMPLES

Four measured sides/corners; or one missing wall backed by aligned rails and a blank interior; or a materially large pitch-black region on a page that passes the dense-ink gate.

REJECTION EXAMPLES

Visible prose crosses the interior; letters impersonate vertical walls; a strip follows the scanner frame; dimensions are implausible at the measured text scale; or stronger geometry already covers it.

GROUPING EXAMPLES

Measured contact or line-end to next-line flow can join components. Indentation, headers, visible text, side columns, and forks split them. Grouping never redraws a polygon.

AUDIT PRINCIPLE

Proposal counts measure search effort. Only the final green polygons are detections. Every saved component retains the route that proposed it, while diagnostic counters preserve both accepted and rejected activity.

Exhibit A: Broken rails, stepped shapes, and safe consolidation

Earlier release | page 3 | 8 components

Approved For Release 2008/12/16 : CIA-RDP79T00975A002900310001-5

25X1

THE POLISH ELECTIONS

25X1

25X1

Faced with no practical alternative, the Polish voters gave overwhelming support on 20 January to Wladislaw Gomulka and to the parliamentary candidates favored by his regime. The mandate apparently exceeded regime expectations, and will strengthen Gomulka's hand in dealing with disruptive elements within his party.

Official results will not become known until later in the week, but it appears that, with perhaps a few exceptions of little consequence, the composition of the Sejm will be as planned by the regime with approximately 50 percent of the seats going to Polish United Workers' (Communist) Party candidates, some 35 percent to the two collaborating front parties, and the remainder to nonparty and Catholic representatives. According to press reports, preliminary returns indicate that all of the party leaders who were candidates were elected.

25X5

The high rate of participation, ranging from 91 to 96 percent of eligible voters in areas reporting so far, was encouraged by regime activists, who made personal visits to those who had failed to register.

25X1

25X1

25X1

25X1

Although there are no reports of intimidation or coercion, few of the electorate other than students took advantage of the opportunity to cast their ballot secretly.

25X1

The elections were orderly and quiet.

25X1

No significant unrest was reported, even in the Lublin area where antiregime agitation has been especially strong.

25X1

22 Jan 57

Current Intelligence Bulletin

Page 3

25X1

Approved For Release 2008/12/16 : CIA-RDP79T00975A002900310001-5

Later release | page 4 | 7 components

Approved for Release: 2019/12/04 C03148913

TOP SECRET

THE POLISH ELECTIONS

Comment on:

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The high rate of participation, ranging from 91 to 96 percent of eligible voters in areas reporting so far, was encouraged by regime activists, who made personal visits to those who had failed to register, and by the Catholic Church. Support by the church was probably also decisive in persuading a large majority of voters--reportedly between 60 and 70 percent--to heed a last-minute plea by Gomulka to vote the straight ticket without deleting names of regime-favored candidates. Although there are no reports of intimidation or coercion, few of the electorate other than students took advantage of the opportunity to cast their ballot secretly.

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22 Jan 57

Current Intelligence Bulletin

Page 3

TOP SECRET

Approved for Release: 2019/12/04 C03148913

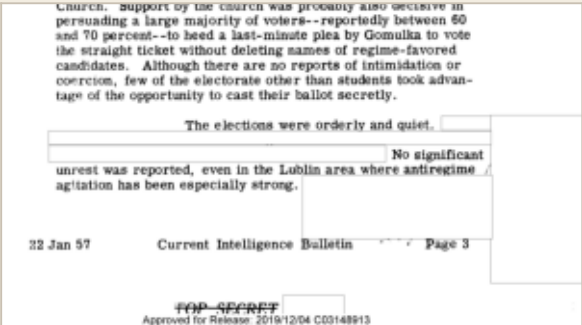
DECISION

Multiple rails and blank slabs are recovered as separate physical components. The text-safe split prevents a broad envelope from swallowing visible prose, and the grouping pass keeps side-column material from bridging into the main paragraph.

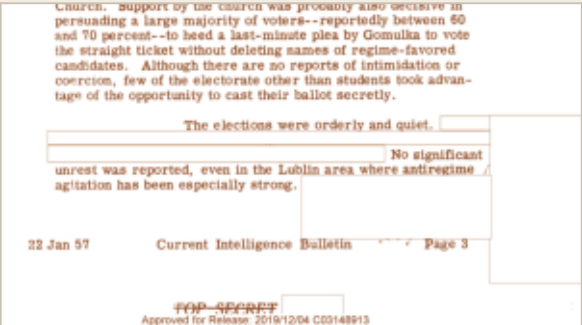
04

Exhibit A: the detector's intermediate view

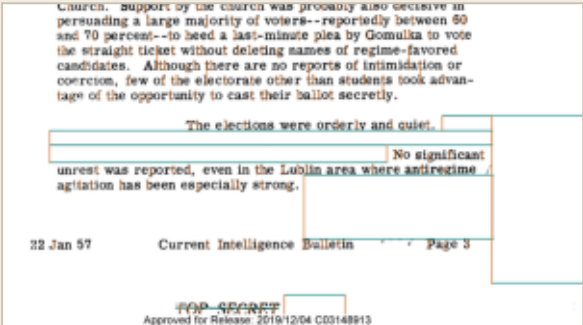
1. Page | Normalize one raster page



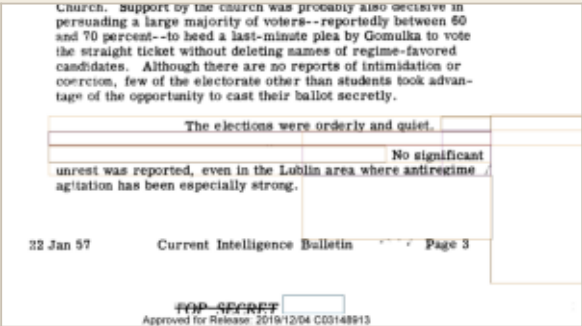
2. Pixel masks | Separate dark and faint evidence



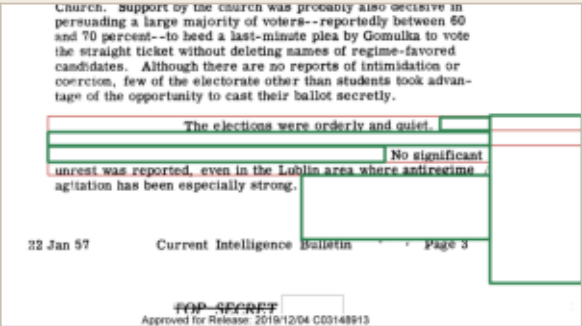
3. Lines | Measure horizontal and vertical support



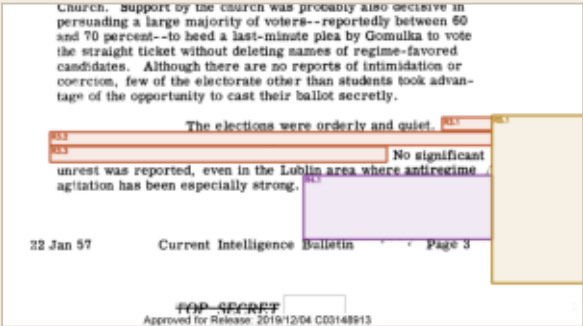
4. Proposals | Propose several physical explanations



5. Guards | Promote measured geometry and block artifacts



6. Consolidate | Freeze components, then infer reading units

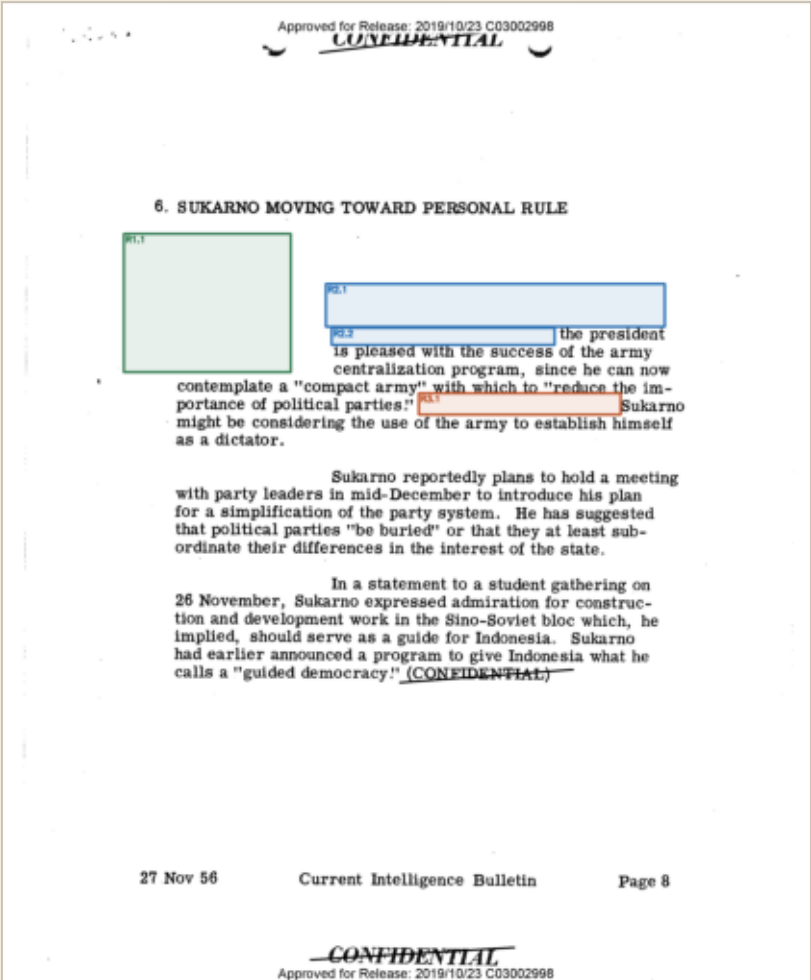
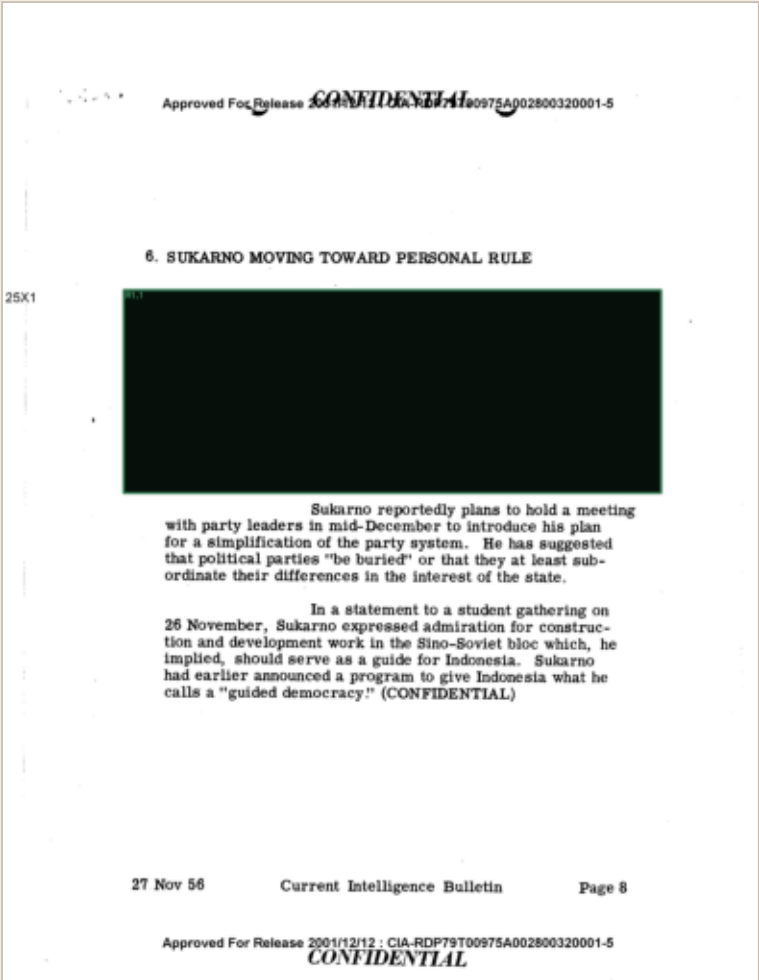


Separate components are retained, then grouped in reading order without filling the visible-text notch.

Exhibit B: A clean blackout and its revealed counterpart

Earlier release | page 9 | 1 components

Later release | page 9 | 4 components

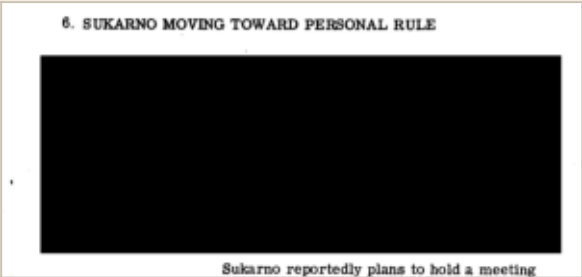


DECISION

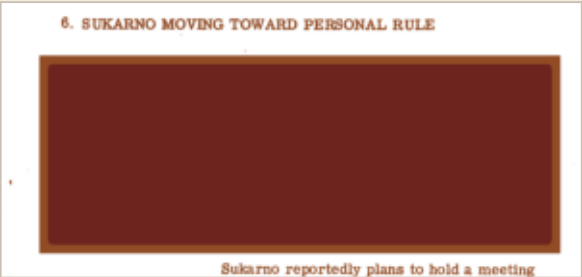
The earlier release contains one materially large, pitch-black rectilinear region, so the page-level blackout route emits one measured component. Only after that geometry is frozen does the audit project the later release's 45 revealed target words back to the earlier scan; all 45 fall within R1.1.

Exhibit B: the detector's intermediate view

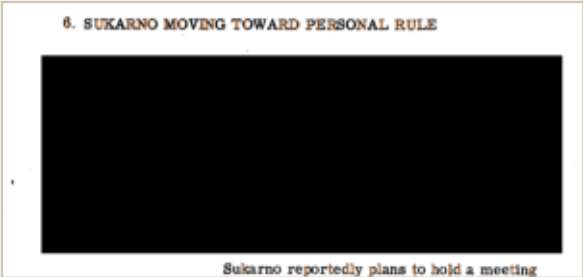
1. Page | Normalize one raster page



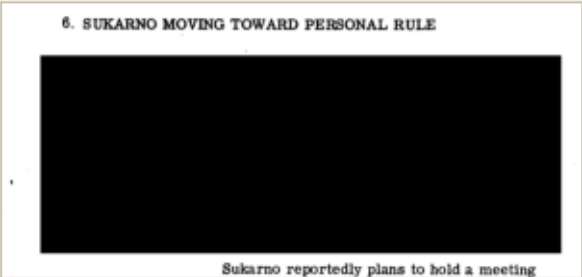
2. Pixel masks | Separate dark and faint evidence



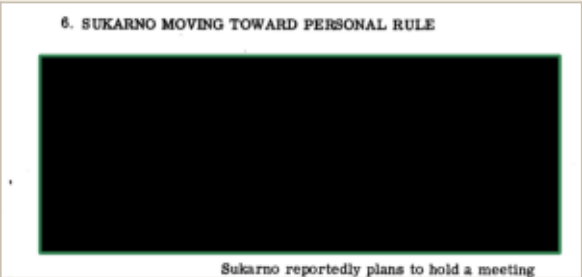
3. Lines | Measure horizontal and vertical support



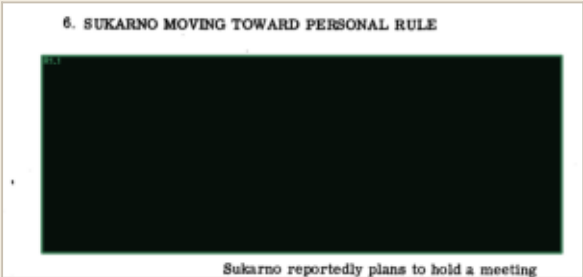
4. Proposals | Propose several physical explanations



5. Guards | Promote measured geometry and block artifacts



6. Consolidate | Freeze components, then infer reading units



The earlier release emits exactly one component and one reading unit. It contains no visible prose.

Exhibit C: Physical overlap before reading-order grouping

Earlier release | page 5 | 8 components

Later release | page 5 | 5 components

25X1A

Approved For Release 2003/10/15 : CIA-RDP79T00975A000700240001-7

4. Chiang Kai-shek plans increased control over military forces:

25X1A

25X1

Chiang Kai-shek plans a reorganization of the military forces to increase his control and that of his elder son, Chiang Ching-kuo. During the past six weeks over 3,000 officers have been processed for retirement, effective in July, and hundreds of promotions are being prepared. Opponents of Chiang like General Sun Li-jen will be deprived of their trusted subordinates.

Chiang has also proposed to reorganize the ground forces into separate area commands, responsible directly to him rather than to the chief of the ground forces.

Comment: Although no effective opposition to Chiang Kai-shek exists among Nationalist leaders, his elder son is disliked by Kuomintang members.

The reorganization of the Nationalist Army, suggested by the US Military Advisory Group, will necessitate some officer transfers. The Generalissimo apparently plans to use this opportunity to retire those officers who oppose him and his son.

5. French report increased Kuomintang activities in Southeast Asia:

25X1A

25X1

the Kuomintang

is multiplying its propagandists and intelligence agents, and attempting to regain control of Chinese schools, unions, and businesses.

- 5 -

25X1A

Approved For Release 2003/10/15 : CIA-RDP79T00975A000700240001-7

Approved for Release: 2019/05/08 C02064591

TOP SECRET

3.5(c)

4. Chiang Kai-shek plans increased control over military forces:

3.3(h)(2)

25X1

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3.3(h)(2)

the Kuomintang's "surge of vitality" through Southeast Asia. It is multiplying its propagandists and intelligence agents, and attempting to regain control of Chinese schools, unions, and businesses.

- 5 -

TOP SECRET

Approved for Release: 2019/05/08 C02064591

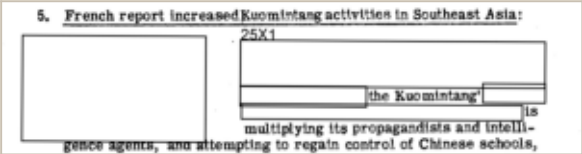
DECISION

The broad left and right masks touch other components around visible words. Geometry is frozen first. Reading-flow logic then joins the true line continuation while keeping the opposite branch in a different R group.

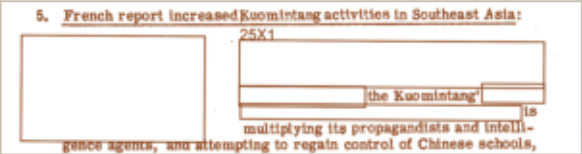
08

Exhibit C: the detector's intermediate view

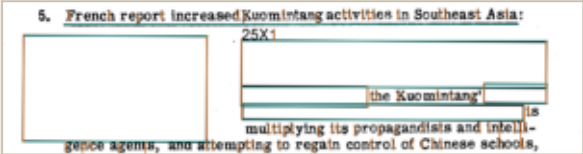
1. Page | Normalize one raster page



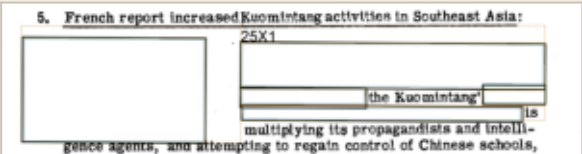
2. Pixel masks | Separate dark and faint evidence



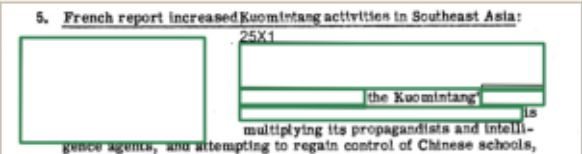
3. Lines | Measure horizontal and vertical support



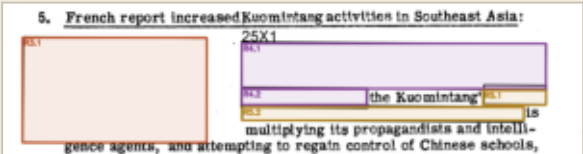
4. Proposals | Propose several physical explanations



5. Guards | Promote measured geometry and block artifacts



6. Consolidate | Freeze components, then infer reading units



R4 follows one line continuation and R5 follows the other; visible words remain outside every polygon.

Astra enters only after detection

1. TARGET

Read the frozen Astra/v4 answer and context.

2. LOCATE

Find exact word polygons in the later PDF text layer.

3. REGISTER

Estimate a SIFT/RANSAC homography between scans.

4. PROJECT

Map later word polygons into earlier-page coordinates.

5. SCORE

Measure word-center coverage against frozen CV polygons.

THE INFORMATION BOUNDARY

The detector sees one raster page and finishes first. Astra supplies semantic target text but no pixel coordinates. The audit obtains coordinates from the later PDF text layer, registers the scans, and compares those projected targets with immutable CV polygons.

STATUS RULES

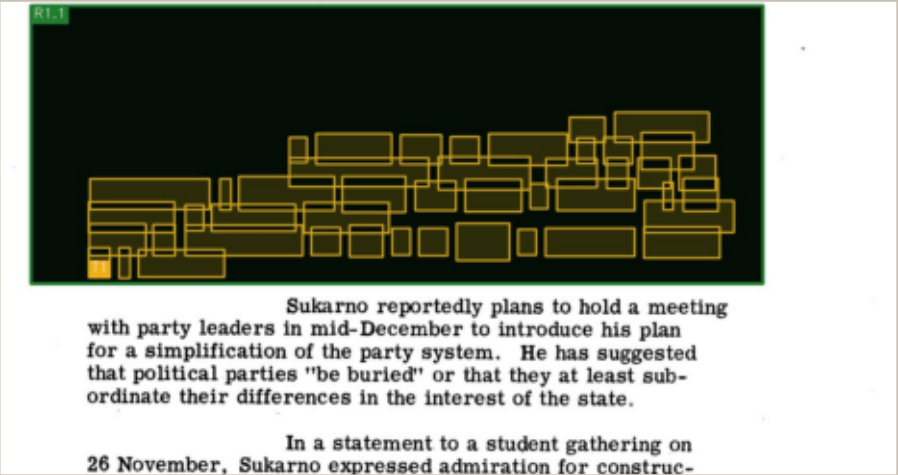
ASSIGNED_FULL: every fragment covers at least 80% of target-word centers. ASSIGNED_PARTIAL begins at 30%. Localization and registration failures remain separate statuses rather than geometry failures.

COMPLEMENTARY MANUAL TEST

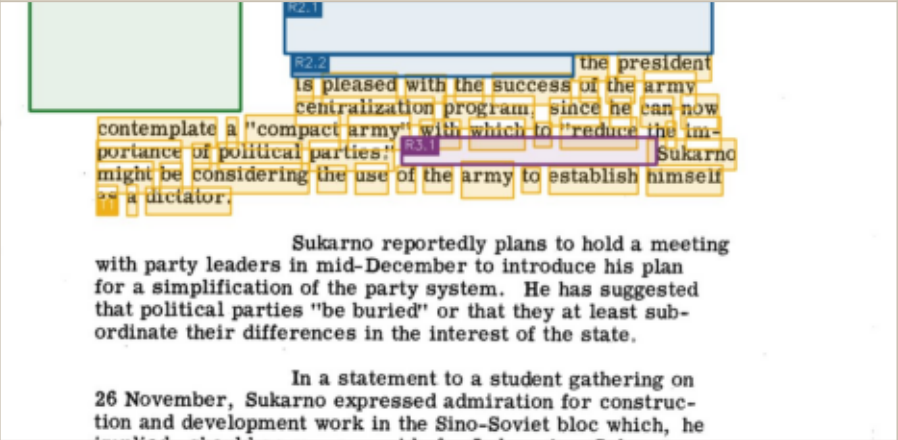
Human polygons are compared with CV polygons after detection using one-to-one component IoU, region matching, and page-union IoU. Astra coverage tests known hidden words; manual gold tests the physical geometry. Both are required for a complete diagnosis.

Worked audit: 45 revealed words, one CV component

Earlier: R1.1 with projected T1 word boxes



Later: target words localized in the PDF text layer



45/45	100%	1,569	0.119px
WORDS COVERED	TARGET COVERAGE	RANSAC INLIERS	MEDIAN REPROJECTION ERROR

SAVED ASTRA/V4 TARGET

[STILL REDACTED] the president is pleased with the success of the army centralization program, since he can now contemplate a "compact army" with which to "reduce the importance of political parties." [STILL REDACTED] Sukarno might be considering the use of the army to establish himself as a dictator.

Interpretation: all 45 localized words are geometrically covered; placeholder markers are excluded. The detector did not transcribe the target.

Output contract and repository use

SAVED GEOMETRY

Every component stores its polygon, bounds, proposal source, and score. A region stores the ordered component set. The JSON therefore distinguishes physical masks from reading units.

POST-HOC AUDIT

The v4 adapter locates saved answer text in the later PDF, registers both scans, projects target words, and measures coverage. It cannot add, remove, or move detector polygons.

RUN

From the repository root:
`python -m box_scripts.v4_audit --plan`
`python -m box_scripts.v4_audit --workers 6 --overwrite`

Open `box_results/items_v4/index.html` for the searchable audit.

INTERPRETATION LIMITS

A detected box is geometry, not recovered language. `ASSIGNED_FULL` means the known v4 target is covered at the configured threshold; it is not a claim that every box on that page has exhaustive gold annotation. Manual labels remain evaluation-only.